

2D Rigid Body Simulator Report

A circle falling under gravity and bouncing against a floor and two side walls

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Coding tool: Codex CLI / VS Code (codex-tui 0.157.0) Model: gpt-6-astra

This report documents the reviewed implementation, numerical validation, and two deliberately injected defects. Production baseline: 71834cd. Local tests were executed by the student; the agent reviewed code and saved evidence.

Part 1 Planning and review

Original agent plan reference

- 1. The original agent plan is reproduced verbatim once in Appendix 1 at the end of this report. The following sections record the requested revisions and the actual implementation.***

Change 1 Development and git sequence

Requested change: Integrate the requested revisions before implementation and replace the later revision-only milestones with substantive development milestones. Show diffs and verification results for my review before accepting each stage.

Reason: Planning review must occur before building. Each commit should represent a meaningful result and provide a recoverable baseline for the next task.

Record: docs/change_log.md and the preserved agent transcript. The final workflow used substantive stages for documentation/CLI, integration, contacts, validation, rendering, and final evidence. Each stage was reviewed before acceptance.

Change 2 Visual diagnosis of contact direction

Requested change: Add a headless plot of vertical velocity before and after resolving floor penetration while the circle is moving upward, comparing correct and deliberately faulty behavior.

Reason: Free fall does not exercise contacts, and energy alone does not reveal velocity direction. This diagnostic can expose an incorrect reversal of outward velocity.

Part 1 Final module layout

File	Responsibility
main.py	Real-time entry point and validated parameter dispatch.
make_figures.py	Run all numerical checks headlessly and save plots.
state.py	SI physics parameters, box geometry, and circle state.
integration.py	Semi-implicit Euler and complete physics-step composition.
contacts.py	Penetration correction, inward-only restitution, and contact records.
rendering.py	PyVista meshes, camera, colors, timer, and fixed-step scheduling.
cli.py	Shared command-line parsing and input validation.
validation.py	Numerical experiments, assertions, energy accounting, and figures.
check_cli.py	Entry-point and parameter regression tests.
check_integration.py	Independent integration, free-fall, and refinement checks.
check_contacts.py	Floor, wall, corner, and motion-direction checks.
check_scheduling.py	Fixed-step timing and overload checks.
final_verify.py	Capture final regressions, figure generation, versions, and exit codes.
build_report.py	Generate PDF report and verify embedded original plan.
preserve_transcript.py	Preserve actual session export bytes and metadata.
package_submission.py	Package submission with complete .git history.
requirements.txt	Full dependency entry point.
requirements-figures.txt	Headless plotting dependencies.
requirements-rendering.txt	PyVista rendering dependencies.
requirements-report.txt	Report generation and PDF verification dependencies.
README.md	Setup, commands, controls, and reproducibility.
.gitignore	Exclude caches, temporary files, and experiment worktrees.
docs/original_plan.md	Unchanged original plan.
docs/change_log.md	The two requested revisions and reasons.
docs/provenance.md	Verified tool/model identity and evidence limitations.
docs/bug_cases.md	Hypotheses, measured defects, causes, and prevention.
docs/report.md	Editable report narrative.

File	Responsibility
docs/report_review.json	Report review status.
docs/submission_audit.md	Requirement audit and evidence references.
docs/transcript/README.md	Transcript and archive instructions.
docs/transcript/manifest.json	Transcript source and coverage metadata.
docs/transcript/*.jsonl	Byte-preserved actual agent conversation snapshot.
figures/*.png	Four required validation figures.
figures/results.json	Measured values, tolerances, versions, and outcomes.
figures/regression_checks.txt	Saved regression output.
evidence/bug_a/*	Actual explicit-Euler experiment logs, diff, values, and plot.
evidence/bug_b/*	Actual unconditional-restitution experiment evidence.
evidence/final/*	Final test logs, figures, versions, and summary.
evidence/stage6_review.diff	Final documentation review diff.
report.pdf	Generated submission report.

Final plan approval

- The approved revisions were incorporated before implementation. The original plan remains unchanged in Appendix 1; docs/change_log.md records both revisions. The reviewed commit history follows substantive development tasks.***

Part 2 Implementation and review

Model and execution

Defaults: $r = 0.2$ m; $m = 1.0$ kg; $g = 9.81$ m/s²; $e = 0.8$; box = 4×3 m; initial position = (1.0, 2.0) m; initial velocity = (1.5, 0.0) m/s; $\Delta t = 1/240$ s.

Each step updates v_y with gravity, updates x and y with the new velocity, increments `step_count` once, and resolves floor and side-wall contacts. Time is `step_count` $\times$ Δt . Penetration is corrected even during outward motion; only inward normal velocity is reflected and scaled by e . Corner components are independent. A fixed orthographic camera looks along z at the x - y plane. A monotonic-clock accumulator advances fixed Δt steps, with at most 60 steps per display callback; discarded backlog is recorded.

Commands and observed results

Command	Actual outcome
python main.py	GUI reported working correctly by the student. Detailed parameter and shutdown observations were not independently recorded.
python make_figures.py	Final local run exited 0; all numerical checks passed and four plots were saved.
python main.py --restitution 1 --dt 0.004166666666666667	Supported and validated options; a separate GUI result for this command is not documented.

Git milestones and review evidence

I reviewed stage diffs and supplied local test outputs before accepting the implementation milestones. The following commits are present in the repository history.

Commit	Meaningful result	Review and verification
14da723	Preserved plan, provenance, and development workflow	Original plan and requested revisions reviewed.
94e2e4f	Validated CLI scaffold	Five CLI tests passed locally.
1c9fdce	State and semi-implicit Euler	Five integration tests passed; signed error and refinement checked.
a5f5654	Floor and side-wall contacts	Six contact tests passed; inward/outward responses checked.
9e0e842	Headless validation and figures	26 numerical checks and 11 regressions passed; figures reviewed.
38cd01e	Dependency organization	Dependency diff reviewed and committed.
71834cd	PyVista rendering and scheduling	Scheduling/CLI checks passed in final rerun; GUI reported working.

Submission references

Repository: D:\CodexCLI\vibecoding\HW2; remote https://github.com/happygary-12321/vibecoding_2.git.

Submission archive must include .git and both local bug branches.

Transcript: docs/transcript/snapshot-20261002T0932437266802Z-rollout-2026-10-01T21-07-30-01a0fa26-efc6-7f61-9193-fb16ebf88047.jsonl. Complete final-session coverage remains to be confirmed.

Environment: Python 3.12.10; PyVista 0.49.0; VTK 9.7.1; Matplotlib 3.10.7; NumPy 2.3.5; ReportLab 5.0.1; pypdf 6.19.0.

Part 3 Simulation validation

Free fall and timestep refinement

The contact-free drop starts at $y_0=10$ m, $v_x=v_y=0$, with $g=9.81$ m/s² and distant surfaces. Runs last 1 s at $\Delta t=1/240$ s and $\Delta t/2=1/480$ s; coarse samples are compared with every second fine sample.

Analytical height: $y(t) = y_0 - \frac{1}{2}gt^2$. Expected signed error: $y_{\text{numeric}} - y_{\text{exact}} = -\frac{1}{2}g\Delta t \cdot t$. Halving Δt should halve the error at matching nonzero times.

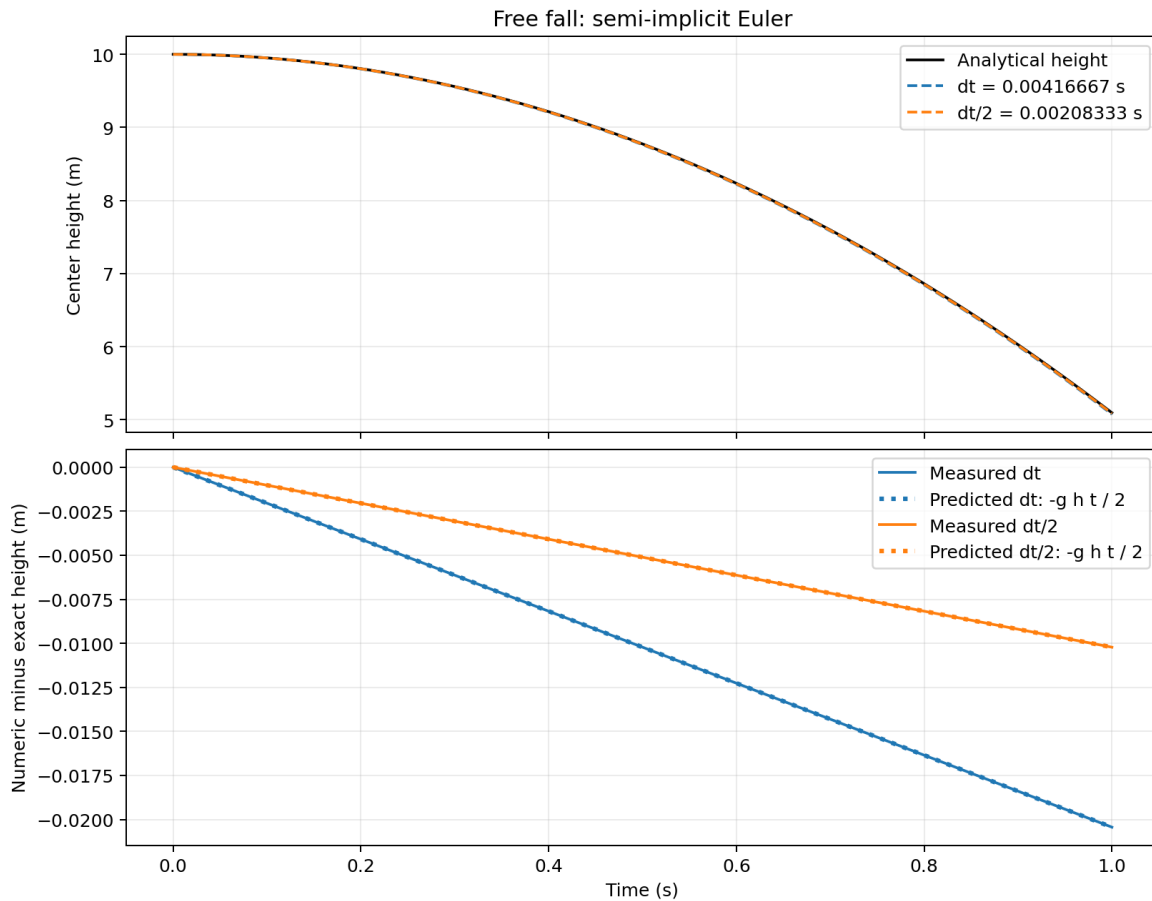


Figure 1. Height and signed error for both timesteps. At 1 s, the exact height is 5.095 m; halving the timestep halves the error magnitude.

Metric at 1 s	Expected	Measured
Height with Δt	5.0745625 m	5.074562500000006 m
Height with $\Delta t/2$	5.08478125 m	5.084781250000005 m
Signed error with Δt	-0.0204375 m	-0.02043749999999367 m
Error magnitude ratio	2	2 within tolerance

The fine signed error is -0.010218749999994614 m. Maximum signed-error residuals are approximately 6.33×10^{-15} m and 8.24×10^{-15} m, below the 10^{-10} m tolerance. Both final vertical velocities are approximately -9.81 m/s. All checks passed.

Total mechanical energy

$E = \frac{1}{2}m(v_x^2 + v_y^2) + mgy$. Between contact-free steps, expected drift is $-\frac{1}{2}mg^2\Delta t^2$ per step. Restitution removes normal kinetic energy at impacts; floor penetration correction can also change potential energy.

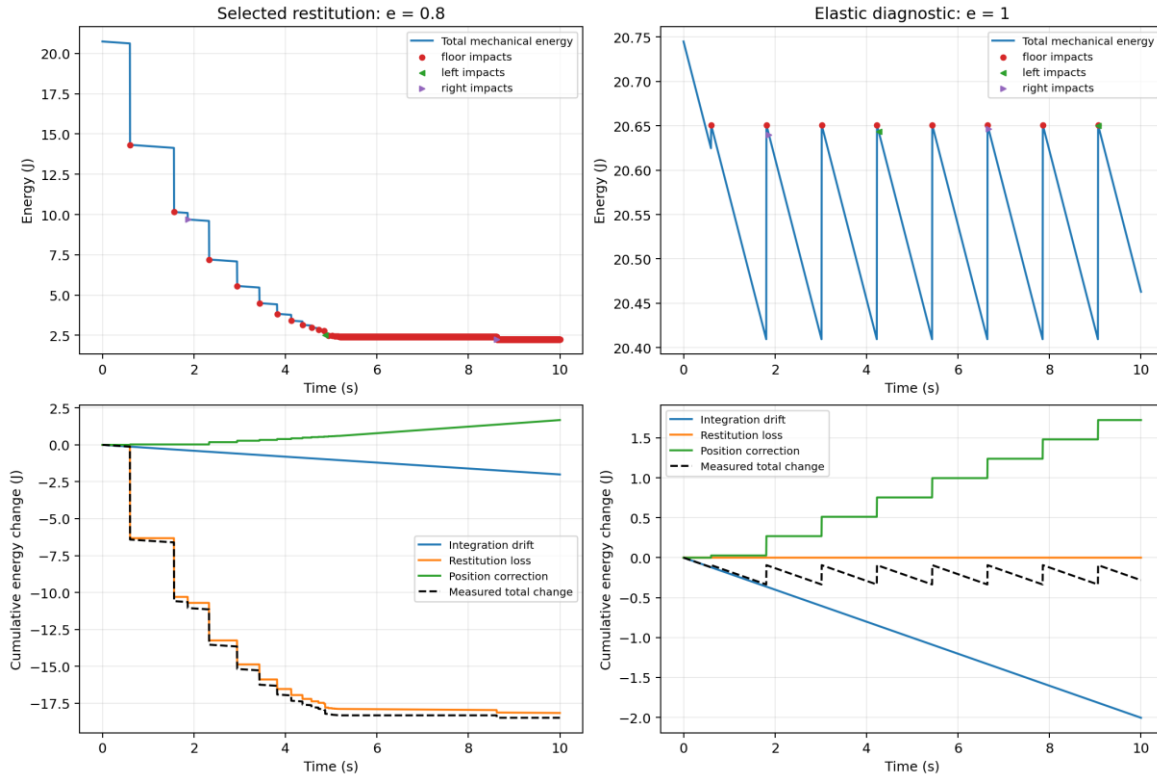
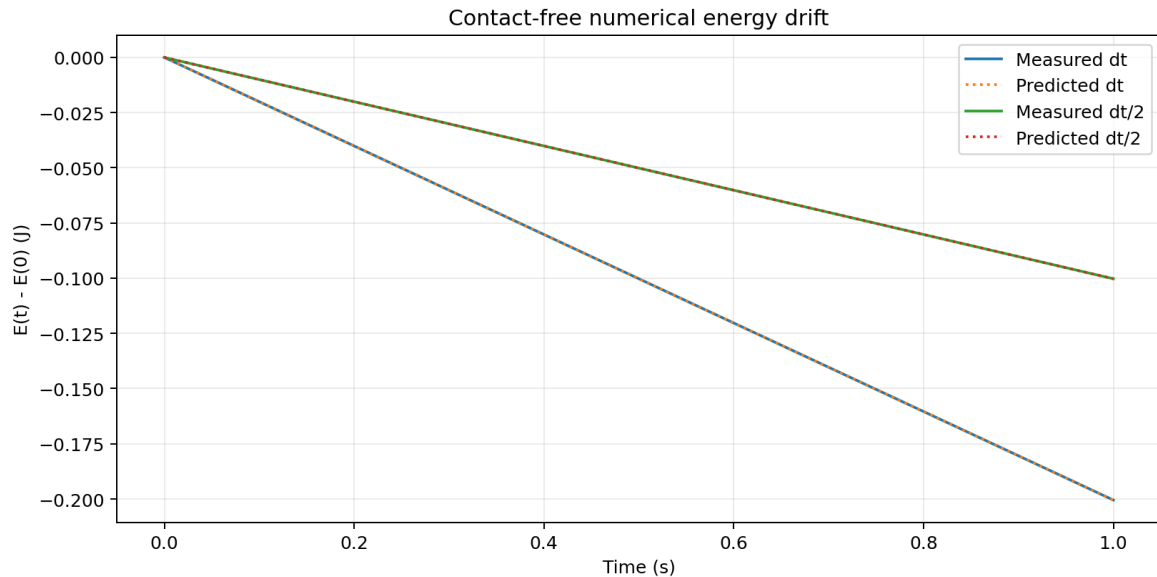


Figure 2. Energy for 10 s at $\Delta t=1/240$ s with $e=0.8$ and $e=1$. Impact markers correspond to actual velocity-changing contacts; cumulative terms distinguish numerical drift, restitution loss, and position correction.

Contact-free energy drift at 1 s is -0.200491875 J for Δt and -0.1002459375 J for $\Delta t/2$; residuals are below 3.22×10^{-13} J against a 10^{-9} J tolerance. At impacts, restitution removes $\frac{1}{2}m(1-e^2)v_n^2$; floor correction adds $mg\Delta y$. Total accounting residuals are 7.40×10^{-13} J and 8.26×10^{-14} J. Elastic restitution removes no kinetic energy, but integration and correction artifacts remain. Small late floor bounces persist because no resting threshold is used.



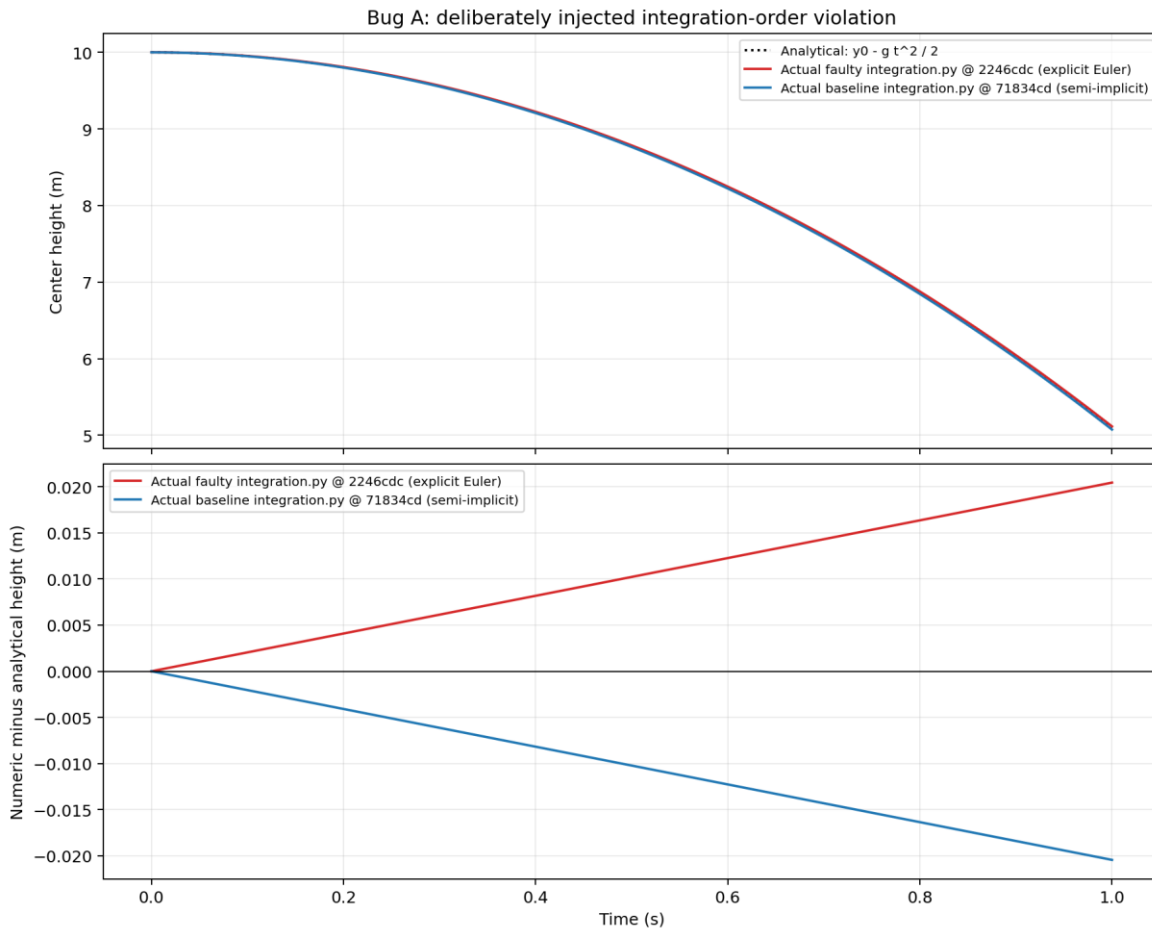
Part 4 Bug evidence and prevention

No adequately evidenced historical physics defects were found. Two defects were deliberately injected on separate local branches from 71834cd. Hypotheses were recorded before mutation; both faulty suites failed while correct-main reruns passed. Main physics remained unchanged.

Bug 1 Explicit Euler

Type: Deliberately injected. Branch: bug/explicit-euler. Commit: 2246cdccb68777de4f97c8800debd2ecff9ab82c.

Symptom: python check_integration.py on the faulty branch exited 1, with two of five tests failing. At 1 s, faulty height was 5.115437500000006 m and signed error +0.020437500000006104 m; the correct height was 5.074562500000006 m with negative error. Both velocities remained approximately -9.81 m/s.



Hypotheses: Before mutation, I expected position-first integration to use the old velocity and produce explicit Euler, reversing the free-fall error sign. Predicted faulty height was 5.1154375 m and error +0.0204375 m. The pre-mutation record is retained in docs/bug_cases.md and the transcript.

Root cause: Updating position before gravity uses the previous vertical velocity. Changing this order changes the numerical method even when gravity and timestep are identical.

Prevention: The independent one-step position check and signed free-fall error assertion detect the wrong order. The correct-main rerun passed five tests with exit 0. The error panel makes the opposite signs explicit.

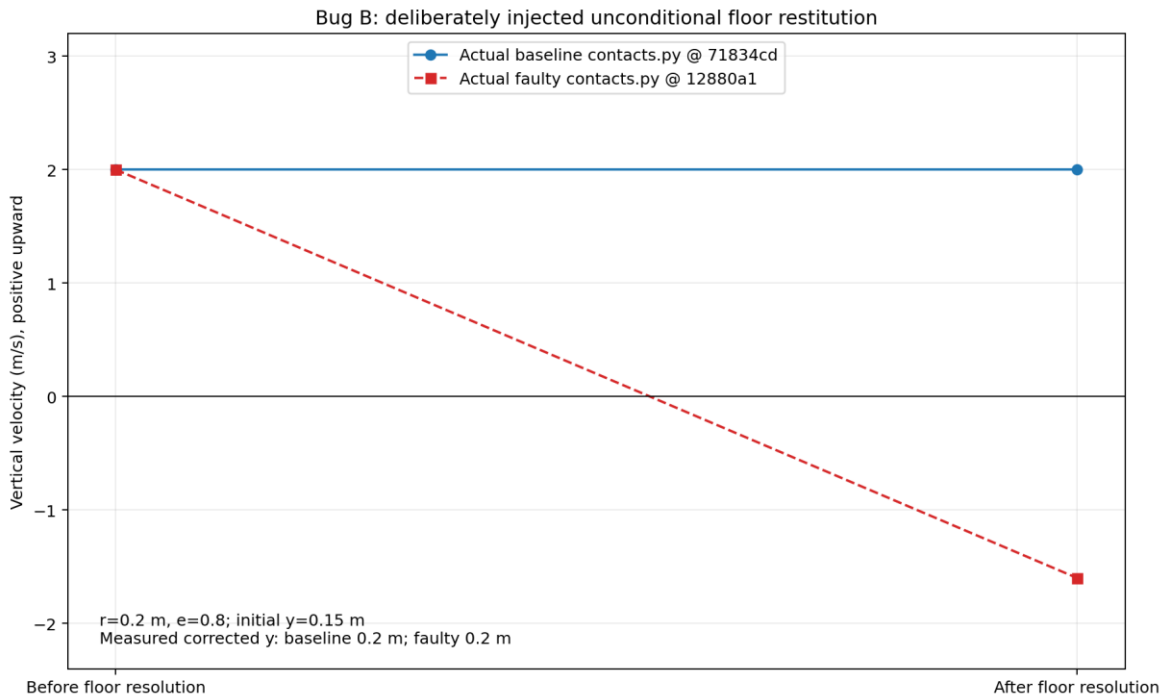
Detecting figure: evidence/bug_a/explicit_euler_comparison.png. Checks: one-step position and free-fall signed error. Clean baseline: 71834cd.

Bug 2 Unconditional floor restitution

Type: Deliberately injected. Branch: bug/unconditional-restitution. Commit:

12880a10f2640f74404fca08491608bf94809d1a.

Symptom: The faulty contact suite exited 1 with two failing outward-motion subcases. For $r=0.2$ m, $y=0.15$ m, $v_y=+2$ m/s and $e=0.8$, both versions corrected y to 0.2 m. Correct v_y remained $+2$ m/s with `is_impact=False`; faulty v_y became -1.6 m/s with `is_impact=True`.



Hypotheses: Before mutation, I predicted that removing the inward-motion condition would reflect an already upward velocity. The controlled prediction was correct $v_y=+2$ m/s versus faulty $v_y=-1.6$ m/s; both positions would be corrected. See docs/bug_cases.md and the transcript.

Root cause: Overlap requires position correction, but it does not always require a velocity response. The faulty rule treats outward motion as an impact and reverses it toward the floor.

Prevention: Contact regressions cover penetrating and exactly touching states moving inward, outward, and tangentially. The signed-velocity plot exposes direction directly. The correct-main contact rerun passed six tests, exit 0; integration and contacts source diffs were empty.

Detecting figure: evidence/bug_b/unconditional_restitution_comparison.png. Check: outward floor motion regression. Clean baseline: 71834cd.

Contact direction diagnostic interpretation

The signed-velocity diagnostic detects Bug 2 because it distinguishes $+2$ m/s from -1.6 m/s. Kinetic energy alone squares velocity and cannot uniquely identify direction. The actual faulty-branch plot above is the injected-bug evidence; the standard headless comparison is labeled as an illustration.

Part 5 Reflection and final submission

Prompting strategy

Reflection draft for review: I used smaller staged prompts after the initial planning prompt, specifying module boundaries, update order, collision direction, and verification requirements. Plan and review records were kept in docs/original_plan.md and docs/change_log.md. Use of a separate project instruction file is not confirmed in the supplied PDF.

Agent performance and my review

Reflection draft for review: The agent organized the modules and supplied testable numerical predictions. I reviewed diffs, ran Python and Git locally, and returned the outputs before accepting each stage. The explicit-Euler experiment showed that plausible motion and correct final velocity were insufficient; the signed-height assertion detected the wrong method. Agent-side interpreter and Git restrictions required local execution.

My implementation choices

Reflection draft for review: The separate physics modules and fixed-step integration provide a useful base for later assignments. Signed-error and contact-direction checks provide stronger evidence than animation alone. Whether I would personally choose another architecture or rendering approach is not stated in the supplied PDF and remains for my own review.

Final submission check

- ✓ Original plan reproduced verbatim once in Appendix 1; two actual changes and reasons included.
- ✓ State, integration, contacts, and rendering have separate modules.
- ✓ Both entry points implemented; restitution and timestep options validated. GUI functionality reported by the student.
- ✓ Validation figures and measured results included; final local verification exited 0.
- ✓ Two deliberate bug cases include symptoms, prior hypotheses, causes, and prevention.
- ✓ At least six meaningful commits confirmed; both local bug commits exist. Final archive inclusion of .git remains to be checked.
- ✓ Codex and gpt-6-astra provenance recorded. Transcript snapshot exists; final coverage remains to be confirmed.

Final local PDF generation exited 0 (16 pages). This completed Word document requires its own final export and visual review.

Remaining submission actions: review personal reflection, refresh complete transcript, and inspect the final archive. Do not mark these actions complete without evidence.

Appendix 1 Original agent plan

The following text reproduces the original plan source verbatim. Markdown characters are retained to preserve the original wording and equations.

This is the proposed original implementation plan. No files will be created or modified, and no implementation will begin until you review it.

1. **Proposed file and module layout**

```
```text
```

main.py — Parse options and run the real-time simulation through the rendering module.

make\_figures.py — Run deterministic checks headlessly and save validation plots and numerical results.

state.py — Define physics parameters, box geometry, and mutable circle state in SI units.

integration.py — Implement semi-implicit Euler integration and compose one complete physics step.

contacts.py — Resolve floor and side-wall contacts, penetration, and restitution.

rendering.py — Own all PyVista objects, camera settings, visual styling, and real-time scheduling.

cli.py — Define shared command-line options and validate their values.

validation.py — Define numerical experiments, assertions, reference values, and plotting routines.

requirements.txt — Record the dependencies needed for rendering, plotting, and report generation.

README.md — Document setup, commands, controls, module responsibilities, and validation outputs.

.gitignore — Exclude caches and reproducible temporary outputs.

docs/original\_plan.md — Preserve this approved original plan verbatim.

docs/change\_log.md — Record your requested changes, your reasons, and their implementation status.

docs/bug\_cases.md — Record real or deliberately introduced bugs and the checks that detect them.

docs/provenance.md — Record the actual coding tool, available model identity, environment, and transcript source.

docs/report.md — Maintain the report text and figure references.

docs/transcript/ — Store complete exported conversation records with an index of their coverage.

figures/ — Store generated validation plots and machine-readable results.

build\_report.py — Convert the report source and preserved plan into report.pdf.

report.pdf — Final submission report, visually inspected after generation.

```
```
```

The physics remains three small modules: state, integration, and contacts. Rendering and validation may depend on physics; physics must never import either.

2. **State representation and module responsibilities**

Use simple dataclasses rather than a general simulation framework:

- `CircleState`: center position `x, y`, velocity `vx, vy`, and integer step count.

- `PhysicsParameters`: radius, mass, gravity magnitude, restitution, and fixed timestep.

- `Box`: width and height, with boundaries at `x = 0`, `x = width`, and `y = 0`.

Defaults are radius `0.2 m`, mass `1.0 kg`, gravity `9.81 m/s<sup>2</sup>`, restitution `0.8`, width `4 m`, height `3 m`, position `(1.0, 2.0) m`, velocity `(1.5, 0.0) m/s`, and timestep `1/240 s`.

Compute simulation time as `step_count × dt`, avoiding cumulative floating-point time addition.

`contacts.py` will optionally return small contact records containing surface name, penetration correction, and normal velocity before and after response. These support validation without adding plotting or rendering concerns to physics.

There will be no rotation, friction, ceiling collision, air resistance, or resting-contact threshold in the initial implementation.

3. **Exact order of operations for one physics step**

For a timestep h :

1. Update velocity: v_x stays unchanged and $v_y \leftarrow v_y - gh$.
2. Update position using the new velocity: $x \leftarrow x + v_x h$, $y \leftarrow y + v_y h$.
3. Resolve floor contact.
4. Resolve left-wall contact.
5. Resolve right-wall contact.
6. Increment the step count and return any contact records.

Contact rules:

| Surface | Contact condition | Position correction | Velocity response |
|------------|-------------------|---------------------|----------------------------------|
| Floor | $y \leq r$ | Set $y = r$ | If $v_y < 0$, set $v_y = -ev_y$ |
| Left wall | $x \leq r$ | Set $x = r$ | If $v_x < 0$, set $v_x = -ev_x$ |
| Right wall | $x \geq W - r$ | Set $x = W - r$ | If $v_x > 0$, set $v_x = -ev_x$ |

Tangential velocity is unchanged. Penetration is corrected even when the circle is already moving away; that outward velocity is preserved.

A floor-and-wall corner contact applies both independent normal responses. Nothing checks the upper boundary.

4. **Real-time rendering strategy**

Render a filled disk centered on the physics position, with visible floor and side walls. Use a fixed orthographic camera looking directly along the z-axis at the x-y plane, with positive y upward. Disable camera interaction that would change this view.

Treat `3 m` as the displayed box height. Show any upper extent marker as a clearly non-colliding dashed guide, subject to your review.

Use a monotonic wall clock and a time accumulator:

- Add elapsed wall time on each display callback.
- Advance physics in fixed increments of `dt` while sufficient accumulated time remains.
- Update the disk actor and render once per display callback.
- Initially display the latest completed physics state without interpolation.

Limit catch-up work to 60 physics steps per display callback. If more complete steps remain after that limit, discard the excess whole-step backlog, preserve the fractional remainder, and record the discarded wall time. The simulation then slows relative to wall time during overload; it never changes the physics timestep or takes a large catch-up step.

Numerical experiments bypass this scheduler and execute an exact number of steps. Closing the window stops the callback and exits cleanly.

5. **CLI parameter validation**

Both entry points will support:

- `--restitution`, default `0.8`.
- `--dt`, default `1/240`.
- Descriptive `--help` output with units and defaults.

`make_figures.py` will additionally accept `--output-dir` and a positive experiment duration.

Reject nonnumeric or nonfinite values, timestep values at or below zero, and restitution outside `[0, 1]`. Reject invalid durations before starting experiments.

Validate the internal configuration as well: positive radius and mass, positive box dimensions, width and height greater than the diameter, finite state values, and nonnegative gravity.

Do not reject an initial center above the displayed height because there is no ceiling. Direct contact tests may intentionally start with penetration.

For fixed-duration comparisons, require duration divided by timestep to be an integer within floating-point rounding tolerance. Explain invalid combinations and suggest an aligned duration; do not silently use a shortened final step. The default validation duration will be `1 s`, giving 240 and 480 steps.

Import PyVista only through the rendering path so `make_figures.py` can run without opening or initializing a GUI.

6. **Numerical experiments, figures, and expected results**

Run every assertion before reporting validation success. Save numerical results with the parameters used. Failed checks produce a nonzero exit status and a clear description.

Free fall and timestep refinement

Use $(y_0 = 10 \text{ m})$, $(v_y = 0)$, $(v_x = 0)$, and a wide, tall box with the circle centered horizontally. Over one second, all contact surfaces remain safely distant.

Run at $(h = 1/240)$ and $(h/2 = 1/480)$. Record the initial sample and every completed step. Compare coarse samples with every second fine sample.

Check:

$$y_{\mathrm{exact}}(t) = y_0 - \frac{1}{2}gt^2$$

$$y_{\mathrm{numeric}}(t) - y_{\mathrm{exact}}(t) = -\frac{1}{2}gt^3$$

At $t=1\text{ s}$, expected values are:

| Quantity | Expected value |
|---------------------------------------|------------------------|
| Exact height | 5.095 m |
| Height at $\Delta t = 1/240\text{ s}$ | 5.0745625 m |
| Signed error at Δt | -0.0204375 m |
| Height at $\Delta t/2$ | 5.08478125 m |
| Signed error at $\Delta t/2$ | -0.01021875 m |
| Final vertical velocity for both runs | -9.81 m/s |

Save a figure containing height versus time and signed error versus time, including the predicted error lines. Verify that halving the timestep halves the error at matching nonzero times.

****Energy between and during impacts****

Define mechanical energy using the circle center:

$$E = \frac{1}{2}m(v_x^2 + v_y^2) + mgy.$$

For a contact-free semi-implicit Euler step:

$$E_{n+1} - E_n = -\frac{1}{2}mg\Delta t^2.$$

Therefore, free-fall energy drift after time t should be:

$$E(t) - E(0) = -\frac{1}{2}mg\Delta t^2.$$

At one second, the default coarse run should lose `0.200491875 J` numerically; the fine run should lose half that amount.

Run the assignment's bouncing configuration for 10 simulated seconds and plot energy versus time, marking floor and side-wall impacts. Also run an elastic diagnostic with restitution `1`.

Record energy at three stages: before integration, after integration but before contacts, and after contact resolution. Account separately for:

- Integration drift.
- Normal kinetic energy removed by restitution:

$$\Delta K = -\frac{1}{2}m(1-e^2)(v_n^-)^2.$$

- Potential-energy change caused by floor penetration correction:

$$\Delta U = mg(y_{\mathrm{corrected}} - y_{\mathrm{penetrating}}).$$

This prevents numerical position correction from being mislabeled as physical impact loss. With `e = 1`, restitution removes no kinetic energy, but timestep and position-correction artifacts can remain.

****Contact checks****

Test contact resolution directly, independently of integration:

- Floor: `y = 0.15 m`, `vy = -2 m/s` becomes `y = 0.2 m`, `vy = +1.6 m/s`.
- Left wall: `x = 0.15 m`, `vx = -2 m/s` becomes `x = 0.2 m`, `vx = +1.6 m/s`.
- Right wall: `x = 3.85 m`, `vx = +2 m/s` becomes `x = 3.8 m`, `vx = -1.6 m/s`.
- Penetrating but moving away: correct position and preserve velocity.
- Exactly touching: reflect only inward velocity; preserve outward and zero normal velocity.
- Corner contact: correct both coordinates and apply restitution to both inward components.
- Restitution `0` and `1`: verify zero and unchanged reflected normal speed respectively.
- Above the displayed box height: verify that no ceiling response occurs.
- Full-step checks: confirm gravity and integration precede contact resolution.

Check that tangential components are unchanged and resolved positions satisfy the allowed floor and wall bounds.

****Tolerances****

For short, direct contact cases, use absolute tolerances of `1e-12` in the relevant SI unit. For the default one-second trajectory and energy identities, use `1e-10 m`, `1e-10 m/s`, and `1e-9 J`.

These tolerances allow floating-point roundoff while remaining many orders of magnitude below the expected discretization errors. Do not require the numerical trajectory itself to match the continuous solution within these tolerances; require its error to match the derived discrete formula.

For substantially different user-selected durations or step counts, document a roundoff allowance scaled by problem magnitude and operation count. Do not loosen tolerances simply to make a failing check pass.

7. ****Phased implementation and meaningful git milestones****

After your review, inspect the workspace and preserve any existing work. Use an existing repository if present; otherwise initialize git before implementation.

Each significant task starts from the previous verified commit. The initial bootstrap necessarily creates the first baseline. Before accepting each subsequent milestone, inspect the diff, run relevant checks, and record their actual outcomes.

| Milestone | Meaningful committed result | Review and verification checkpoint |
|---|---|---|
| 1 | Original plan, provenance notes, project documentation, and executable CLI scaffold | Verify both help commands and honest “not implemented” behavior for unfinished paths |
| 2 | Physics state and semi-implicit integration with initial free-fall checks | Verify velocity, position, sample times, and signed error |
| 3 | Floor and side-wall contact handling with deterministic checks | Verify all response directions, penetration, corners, and restitution endpoints |
| 4 | Complete headless validation entry point and figures | Run all checks; inspect plotted values and energy accounting |
| 5 | PyVista rendering and fixed-step scheduling | Run a GUI smoke test; check camera, geometry, shutdown, and slow-frame behavior |
| 6 | Your first requested revision | Record your reason; review the diff and run affected checks |
| 7 | Your second requested revision | Record your reason; review the diff and run affected checks |
| 8 | Completed bug evidence, report, transcript index, and reproducibility instructions | Run final checks, inspect report pages, review final diff and repository status |

Milestones 6 and 7 depend on actual requests from you. They will not be fabricated to meet the commit count. Substantive fixes may add further commits. No empty commits or cosmetic splitting will be used.

8. ****Bug evidence and deliberate-bug branches****

When a real bug occurs, preserve the failing command, relevant parameters, expected and observed values, diagnostic output, affected commit, fix diff, and successful rerun. Record the explanation while the evidence is available.

Do not retroactively describe an unobserved defect as a real bug.

If fewer than two real bugs occur, create enough deliberately faulty cases on separate branches from a known passing baseline. Candidate faults are:

- Change integration to update position before velocity. The signed free-fall error check should detect the wrong sign.
- Apply restitution to a penetrating circle that is moving away. The outward-contact check should detect an incorrect velocity reversal.

Commit each deliberate defect on its branch, run the existing detecting check, and preserve its nonzero exit result and output. Label it explicitly as an injected bug. Keep the branch and commit reference; do not merge the defect into the submission branch.

Return to the clean baseline and rerun the detecting checks. Each case study will distinguish the failing branch from the correct submission behavior.

9. ****Report and transcript preparation****

Preserve this plan verbatim in `docs/original\_plan.md` after implementation is authorized. Later changes belong in the change log; the original plan is not rewritten.

Preserve complete conversation exports, including prompts, responses, and tool activity to the extent the application export provides them. Keep original exports intact and index their coverage. If complete export requires a manual action from you, identify that requirement early; do not reconstruct missing transcript content or claim a summary is complete.

Record the coding tool from the actual environment and the model identifier only from authoritative session metadata. If the exact model identifier is unavailable, state that explicitly and request the displayed identifier or export metadata instead of inferring it.

The report will contain:

- The original plan verbatim.
- At least two actual changes you requested and their stated reasons.
- Final module layout and step order.
- Validation methods, parameters, figures, actual results, and tolerances.
- Two evidenced bug case studies, labeled real or deliberately injected.
- A short reflection on numerical accuracy, contact artifacts, architecture, and the review process.
- Tool/model provenance, commit references, and transcript location.

Generate `report.pdf`, render its pages for visual inspection, and check equations, figure labels, page breaks, and the full original-plan text before delivery.

10. ****Decisions for your review****

The proposed defaults are:

- Treat the box as open at the top, with a non-colliding upper extent guide.
- Use discrete post-step penetration correction; do not add continuous collision detection.
- Preserve the specified restitution rule without a resting-contact threshold, accepting possible small late-time bounces.
- Cap real-time catch-up at 60 steps per callback and discard excess wall-time backlog.
- Use a one-second free-fall experiment and a ten-second bouncing-energy experiment.
- Keep physics dependencies minimal and use a compact assertion-based validation suite.
- Require two genuine requested revisions before considering the report complete.

Implementation is paused here for your review.